

Admissible Relational Space $\mathfrak{R}_q$

Coordinate-Generation Operators

Representative groups; recursively composable

$\partial, d, \Delta, D^\gamma$ **Local**
Derivatives, differences,
tangent / phase constructions

$\int, *, \mathcal{L}_\tau$ **Nonlocal / History**
Integrals, convolutions,
lags and memory operators

$\mathcal{T}, \mathcal{B}, \Pi$ **Algebraic / Basis**
Transforms, ratios, reductions,
nondimensional groups and bases

$\mathcal{G}, \mathcal{S}$ **Structural / Statistical**
Geometry, symmetry, graph,
spectral and statistical maps

Φ_θ **Learned**
Embeddings, kernels,
neural / surrogate coordinates

$\mathcal{M}$ **Model Primitives**
Numerical simulators, pretrained
models, scientific codes, agents

*Open, Expandable
Discovery Frontier*



$$G = G_{i_k} \circ \dots \circ G_{i_2} \circ G_{i_1}$$

Operators may compose recursively

Key  Category  Illustrative subfamily

Illustrative, non-exhaustive categories; each SIR run instantiates a finite, auditable grammar.

Overlapping categories permit hybrid/composed representations

Symbolic Regression: One Restricted Subfamily

Algebraic /
Functional

$y = F(x)$ $F = 0$
 P/Q $\mathcal{R}_{\text{SR}}$

Differential /
Evolution

d/dt ∂_i
 $D_g^\gamma f$ $\nabla^2 u$

Integral /
Nonlocal / Memory

$\int$ $u(t - \tau)$
 $K * u$

Geometric /
Topological /
Symmetry

κ_γ $d_{\mathcal{M}}$
 $g \cdot x$ H_k

Spectral /
Modal / Multiscale

$L\phi = \lambda\phi$ $\sum a_k \phi_k$
 $u_0 + \varepsilon u_1 + \dots$

Statistical /
Stochastic /
Population

$p(y|x)$ $\theta^{(j)} \sim P_\theta$
 $\mathbb{E}, \text{Cov}$

Constraint /
Conservation /
Variational

$C = 0$ $C \leq 0$
 $\nabla \cdot J$ $\delta J = 0$

Discrete /
Graph / Hybrid

$x_{n+1} = F(x_n)$ $L_G x$
 $R_k|_{\Omega_k}$

Learned /
Nonparametric

$z = \Phi_\theta(x)$ $K(x, x')$
 $\hat{F}_{\text{data}}$